EPSILON SOUNDS · USER MANUAL · V0.2.0

Epsilon Chop

Drop in a sample. Epsilon Chop cuts it into colour-coded chops, lays them on a 16-step ribbon, and flips them with one roll of the dice. This manual covers every control, shows where each one lives, and explains what happens inside.

VST3 · AU · Standalone | Up to 32 chops | 16 steps × 4 banks | MIDI root C5



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04 Arrangement
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CHAPTER 01

Five clicks to your first flip

1. **Drop a sample** anywhere on the window, or click **LOAD SAMPLE**. WAV, AIFF, FLAC, MP3 and OGG, up to 120 seconds.
2. Keep **TRANSIENT** for drums and vocals, or pick **GRID → 16** for a clean loop.
3. Click **IN ORDER**, then **PLAY**. You hear the loop rebuilt from its own chops. FL Studio doesn't need to be playing.
4. Press **RANDOMIZE**. Chops trade places, and the readout under the button tells you which ones, for example `Swapped 11 ⇔ 13`.
5. Keep the result, or click **UNDO** and roll again. Save the ones you like to banks **B**, **C** and **D**.

CHAPTER 02

The header

What's loaded, how loud it is, and where new samples come in.



1 Name
The chop-stripe logo. Every colour in it is a chop colour.

2 Sample pill
Name, length, sample rate, mono/stereo and chop count. Shows *Loading…* and any error, such as an unreadable or too-long file.

3 OUTPUT
−24 to +6 dB, default −3 dB; double-click to reset. Smoothed over 8 ms so it never clicks. Automatable.

4 LOAD SAMPLE
WAV, AIFF, FLAC, MP3 or OGG; mono or stereo, 8–192 kHz, up to 120 s. Dropping a file anywhere, or clicking the empty waveform, also loads.

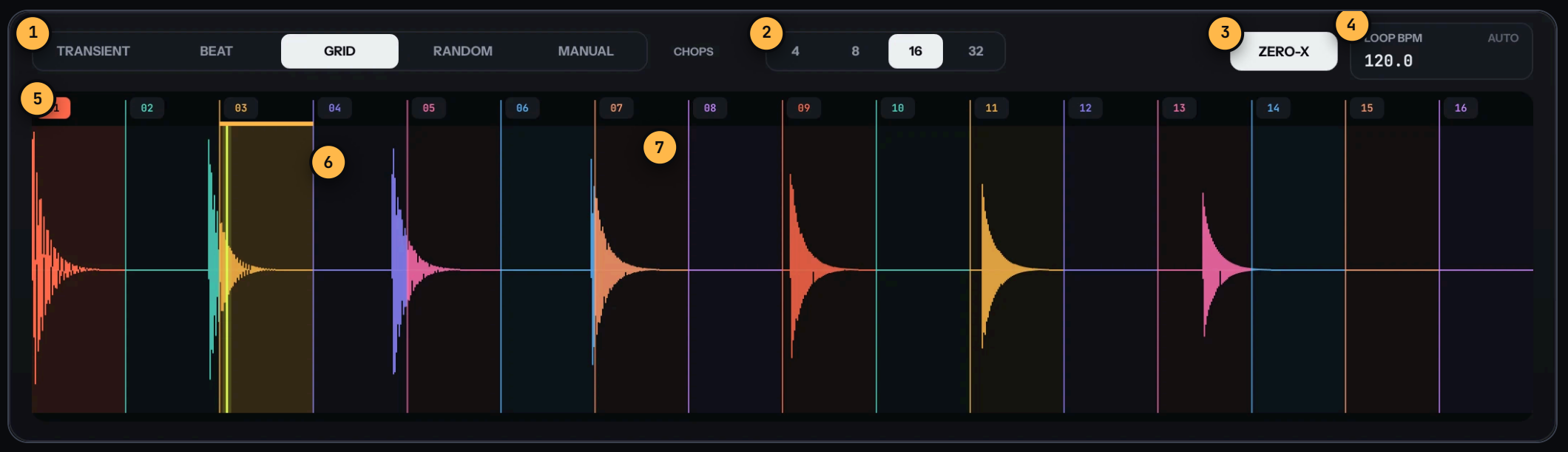
Loading never interrupts audio.

Decoding and analysis run on a background thread, and the new sample crossfades in over 2 ms. Your four banks are kept. Undo history clears, and LOOP BPM goes back to auto-detect.

CHAPTER 03

Chop modes & the waveform

Five ways to cut. Every chop gets its own colour, and that colour follows it everywhere.



1 **Chop mode**
TRANSIENT, BEAT, GRID, RANDOM or MANUAL. The controls to its right change with the mode.

2 **Mode setting**
SENSITIVITY + MAX (Transient), DIVISION (Beat), CHOPS (Grid, Random), RE-CHOP (Random).

3 **ZERO-X**
Snaps every automatic cut to the nearest zero crossing within ±2 ms, so chops start without a click.

4 **LOOP BPM**
Detected tempo of the loop. Drag to correct (Shift = fine), scroll for ±1, double-click for AUTO.

5 **Chop tag**
Number and colour of each chop. The selected chop's tag fills in.

6 **Sounding chop**
Coloured bar plus the volt playhead: exactly where the audio is being read.

7 **Cut lines**
Drag to move (switches to Manual). Double-click empty space to add, right-click a line to remove.

The five modes, on the same drum loop

TRANSIENT



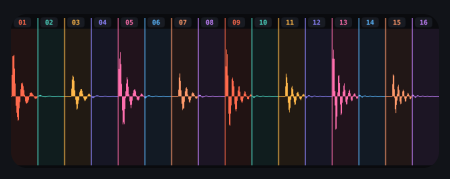
Cuts just before each hit. SENSITIVITY (0–100) sets how quiet a hit can be. MAX (8 / 16 / 32) caps the count, keeping the strongest.

BEAT



Cuts on 1/4, 1/8 or 1/16 of LOOP BPM, nudged onto a nearby hit. Best for loops in time.

GRID



Equal chops: 4, 8, 16 or 32. Clean and predictable for pads and tidy loops.

RANDOM



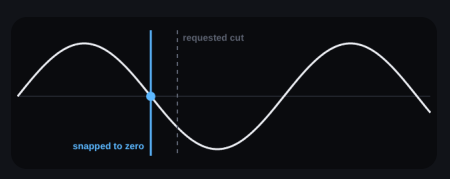
Random cut points for happy accidents. RE-CHOP, or click RANDOM again, rolls new ones.

MANUAL



Your own cuts. Starts from whatever you were looking at. Double-click to add, drag to move, right-click to remove.

ZERO-X



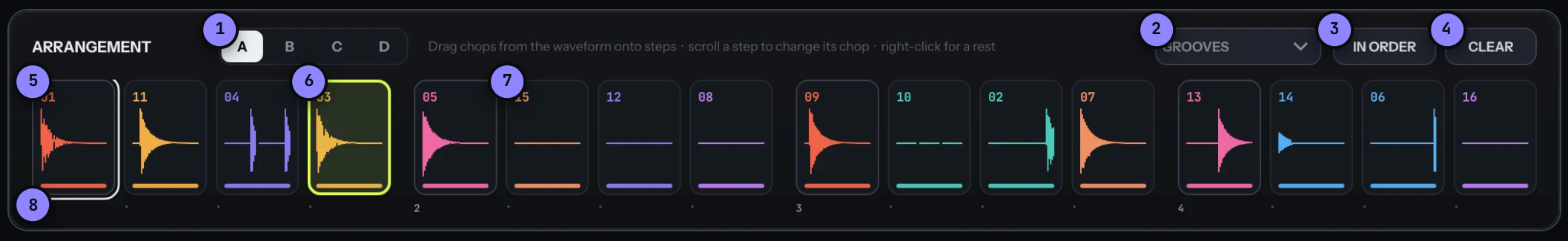
Every automatic cut moves to the nearest zero crossing within ±2 ms. The chop then starts at silence instead of mid-wave, so there's no click.

WAVEFORM ACTION	RESULT
Click a chop	Plays it and selects it. The selected chop is used whenever a rest becomes a hit.
Drag a chop out	Picks it up as a coloured tag. Drop it on any step.
Drag a cut line	Moves the cut, keeping chops at least 4 ms long. In an automatic mode, switches to Manual.
Double-click	Adds a cut (up to 32 chops).
Right-click a line	Removes that cut.

CHAPTER 04

The arrangement ribbon

Sixteen steps. Each one shows the actual audio of the chop it plays.



1 Banks A–D
Four independent 16-step patterns sharing one sample. Switching is instant.

2 GROOVES
Starting rhythms: Straight Eighths, Jersey Bounce, Drill Space, Dembow Pocket.

3 IN ORDER
Lays chops out as recorded, evenly over 16 steps. The best starting point for a flip.

4 CLEAR
Turns every step of this bank into a rest.

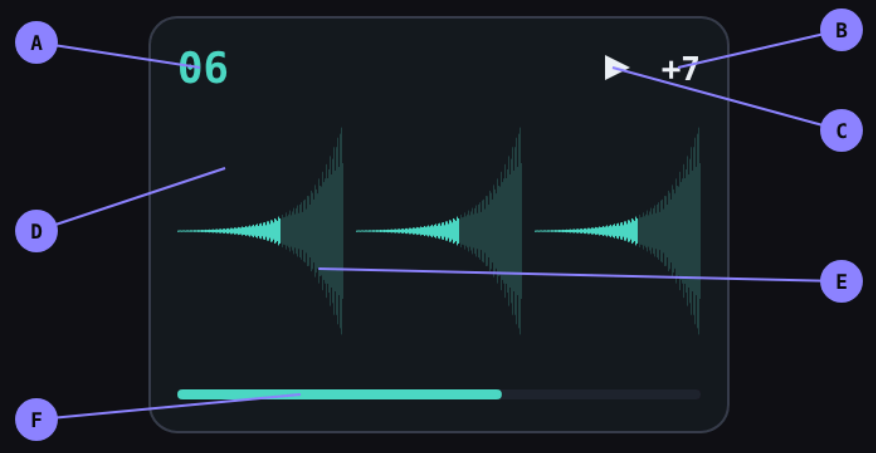
5 Selected step
White outline. The STEP panel edits this one.

6 Playing step
Volt outline: the step sounding now.

7 Ratchet
The chop drawn once per repeat. This step plays four quick hits.

8 Beat ruler
Steps grouped in fours. Beats 1–4 of a bar at RATE 1/16.

Anatomy of a step



A Chop number
In the chop's colour.

B Pitch badge
Semitones, for example `+7`.

C Reverse arrow
The waveform is mirrored too.

D Ratchet copies
One per repeat, 1–4.

E Faded tail
Audio cut off by the gate.

F Gate bar
Length of the hit, 10–100% of the step.

ON A STEP	RESULT
Click	Select it and hear its chop.
Right-click, Shift-click or double-click	Toggle rest ⇔ hit. A new hit uses the selected chop.
Scroll wheel	Step through chops, hearing each one.
Drop a chop	Assign it, keeping the step's gate, ratchet, pitch and reverse.

CHAPTER 05

Transport: play it, sync it, swing it

Hear the arrangement with FL stopped, or lock it to FL's bar.



1 PLAY
Runs the bank on the internal clock from step 1. It lights volt while running. The label reads PREVIEW, PLAYING or HOST.

2 TEMPO
Follows FL (tag HOST). With no host tempo, drag 40–240 BPM; double-click for 120.

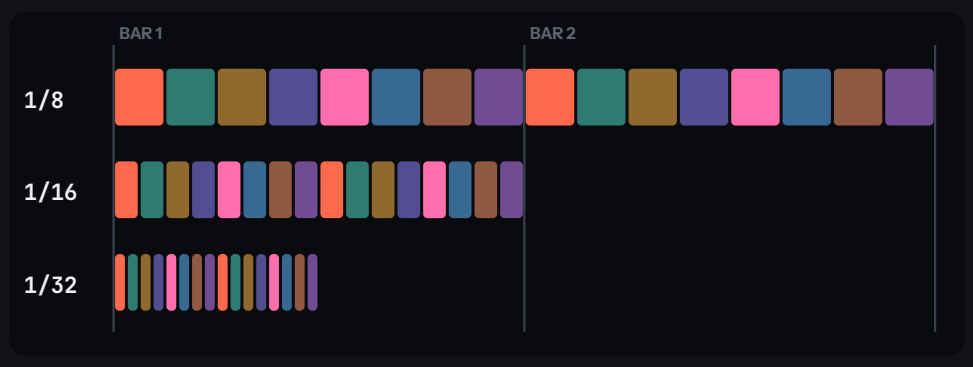
3 RATE
One step = 1/8, 1/16 or 1/32. 16 steps = 2 bars, 1 bar or half a bar.

4 SWING
0–60%. Delays every second step; see the diagram below.

5 HOST SYNC
Plays whenever FL plays, with step 1 locked to the bar line. Stops when FL stops.

6 TEMPO FIT
Repitches the sample so LOOP BPM matches the tempo (¼× to 4×). Pitch moves with speed, like a turntable.

RATE: how far 16 steps reach



SWING: same pair length, later off-step



Rose = first step of each pair · Iris = the swung step

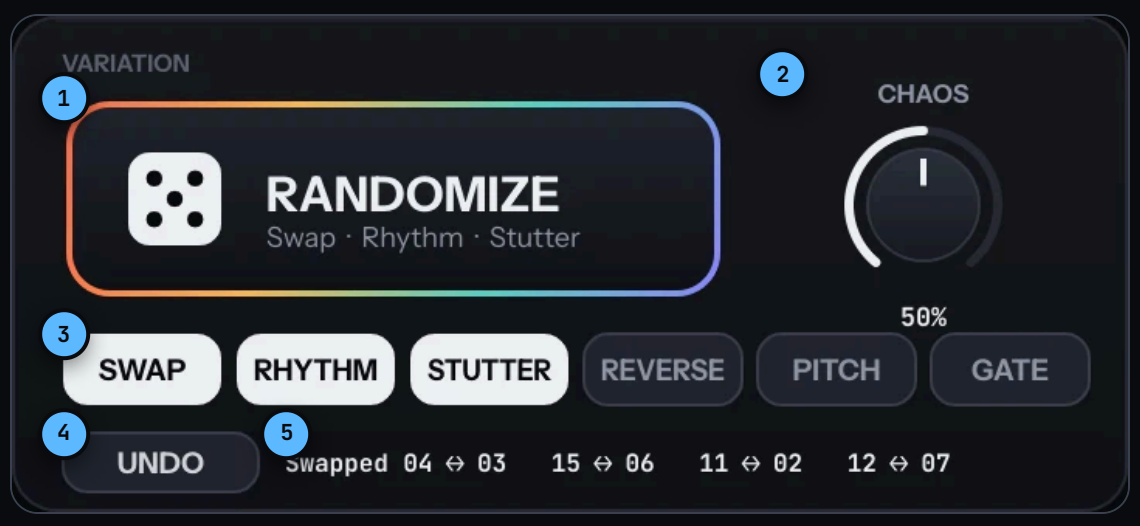
HOST SYNC and PLAY together?

Host sync takes over while FL's transport runs. When FL stops, PLAY carries on with the internal clock.

CHAPTER 06

Randomize: flip it fast

Toggles choose *what* may change. CHAOS chooses *how much*. Every roll can be undone, and if nothing is playing, a roll starts PLAY so you hear it at once.



1 **RANDOMIZE**
Rolls the current bank. The subtitle lists the active toggles.

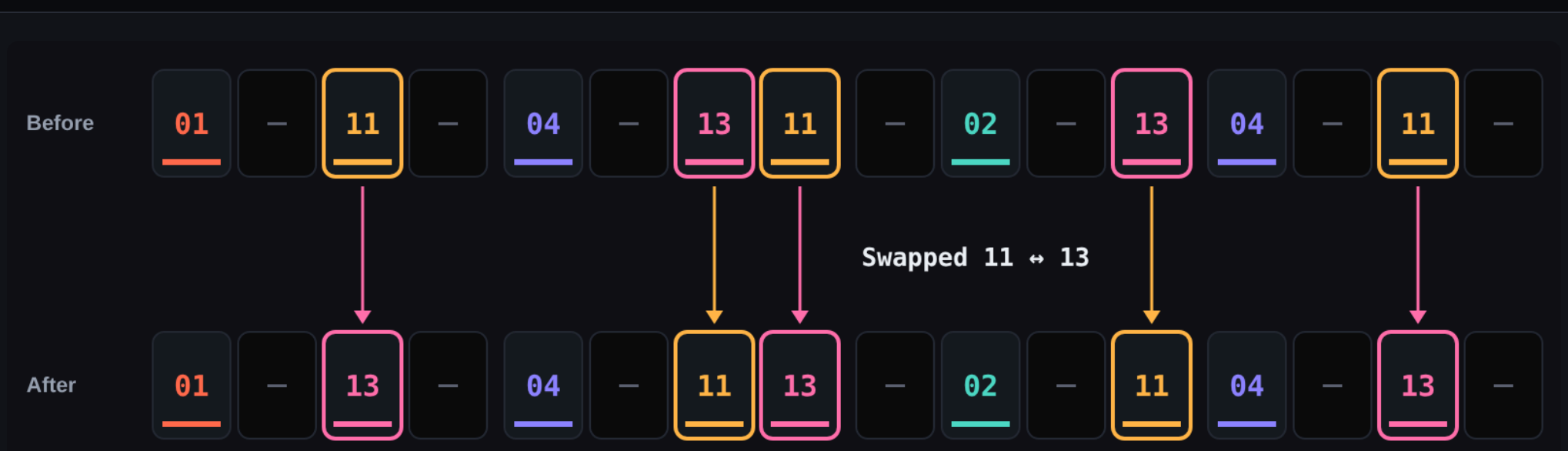
2 **CHAOS**
Size of each roll, 0–100%. The default is 50%.

3 **Toggles**
SWAP, RHYTHM, STUTTER, REVERSE, PITCH, GATE. White = on.

4 **UNDO**
32 levels: cuts, grooves, step edits and rolls.

5 **Swap readout**
Exactly which chops traded places on the last roll.

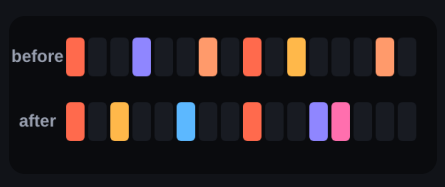
SWAP: chops trade places



A swap of **11** and **13** turns *every* 11 into 13 and every 13 into 11. Timing, gates, ratchets, pitch and reverse stay put. The partner can be any chop in the sample, so swaps also pull in parts of the sample that weren't in the bank yet. Number of swaps = CHAOS × (different chops in the bank) ÷ 2. At 95% and above, every chop is shuffled at once.

The other five toggles

RHYTHM



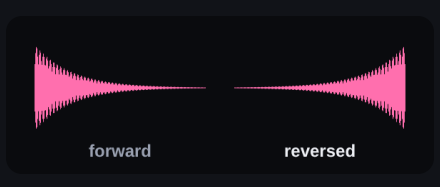
Moves CHAOS × hits to new steps, favouring downbeats (1, then 9, 5, 13). Sometimes it adds or drops a hit, keeping 4–13 hits and never removing step 1.

STUTTER



Ratchets of 2, 3 or 4 repeats inside a step. Each hit has a CHAOS × 32% chance. Repeats play at 82% level.

REVERSE



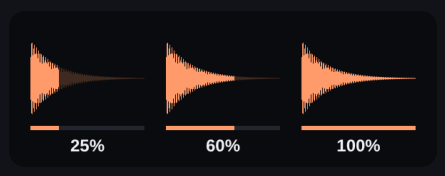
Plays chops backwards. Each hit has a CHAOS × 30% chance.

PITCH



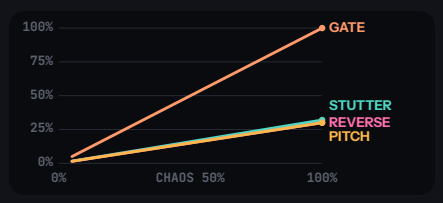
Shifts hits by −12, −7, −5, +5, +7 or +12 semitones. Each hit has a CHAOS × 30% chance.

GATE



Sets hit lengths to 25, 40, 60, 80 or 100%. Each hit has a CHAOS chance.

CHAOS, per hit



How likely each hit is to change at a given CHAOS setting.

Good to know

With STUTTER, REVERSE or PITCH on, every hit is re-rolled for that property, so turn a toggle off to keep what you set by hand. An empty bank gets a new 5–9 hit phrase. A roll that would change nothing forces one swap. Bank switching, the toggles and CHAOS are settings, so UNDO leaves them alone.

CHAPTER 07

The step inspector

Shape the selected step by hand.



1 **Step number**
Which of the 16 steps you're editing.

2 **Chop chip**
The chop it plays, in its colour, or REST.

3 **< >**
Previous or next chop, wrapping around and auditioning each one.

4 **GATE**
10–100% of the step (per repeat with ratchets). A shorter chop ends on its own.

5 **PITCH**
±12 semitones by resampling: up is shorter and brighter. Double-click for 0.

6 **RATCHET**
1–4 equal repeats inside the step.

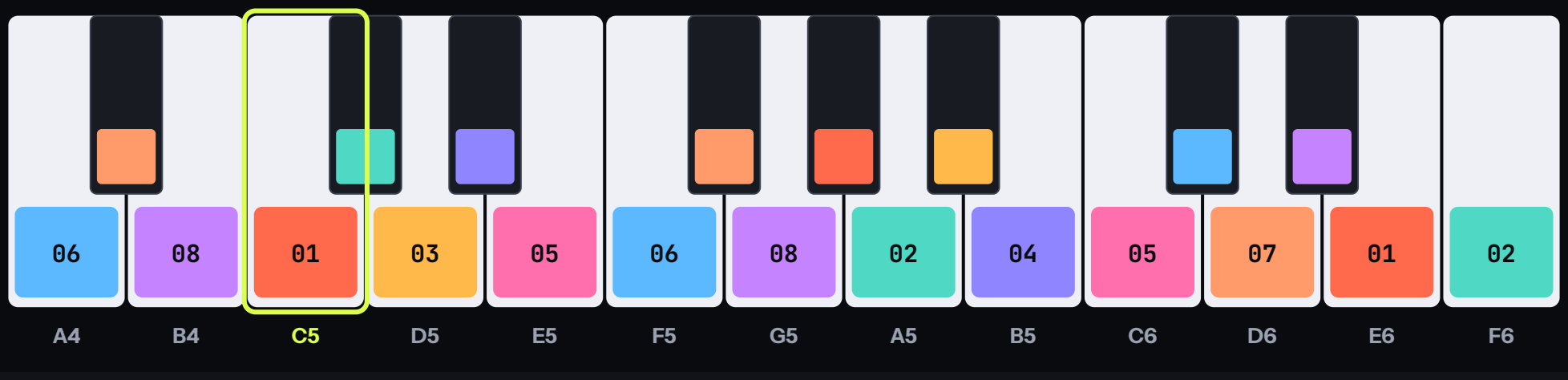
7 **REVERSE**
Plays from the chop's end back to its start.

8 **REST**
Toggles rest ⇔ hit (using the selected chop).

CHAPTER 08

MIDI & the Piano Roll

Every key plays a chop. C5 is where FL's step sequencer plays, so a new channel makes sound straight away.



With 8 chops: **C5 → Chop 01**, each key up is the next chop, G#5 wraps to Chop 01 again, and B4 plays Chop 08. Set the root in the footer, from C1 to C7. Velocity sets the level, and CC 120/123 silences notes.

Mono choke

A new hit fades the previous one out over 2 ms: classic MPC chopping. MIDI hits play the chop plainly. Step pitch, reverse, ratchet and gate belong to the ribbon.

Automation

Host Sync · Output · Step Rate · Swing · Tempo Fit · Root Note · Internal Tempo. Cuts, banks and randomizer settings save with the project.

CHAPTER 09

Under the hood

From audio file to audio out.

BACKGROUND

Decode

64k-frame blocks at the file's own rate. Bad values are zeroed.

BACKGROUND

Onsets

Spectral flux with a 3-bin max filter. Up to 256 hits, kept with their strengths.

INSTANT

Cuts

The mode picks from the stored onsets, then ZERO-X snaps each cut.

LOCK-FREE

Session

An immutable snapshot is published to the audio thread.

AUDIO

Clock

Host PPQ or the internal clock. Sample-exact steps, swing and ratchets.

AUDIO

Voices

Mono choke, 8 voices, 2 ms fades. 32-tap sinc for pitch and rate.

Finding the hits

A Hann window of at least 16 ms hops a quarter window at a time. The log magnitude of the louder channel is taken per bin. Novelty is each bin's rise above the *maximum of its three neighbours* one frame earlier, which ignores vibrato and slides (Böck & Widmer, DAFx 2013). A frame becomes a candidate at a local peak that clears 1.25 × the mean of ±10 frames. The cut then slides back to where the waveform first reaches 15% of its local peak, minus 1 ms. Hits closer than 35 ms to a stronger one, or within 10 ms of either end, are dropped.

Choosing the cuts

Transient keeps onsets whose strength is at least strongest × (0.6 × (1 − sensitivity/100)² + 0.004), and the strongest ones win when there are more than MAX allows. That's why the slider responds instantly. **Beat** divides by LOOP BPM and pulls each cut onto a hit within 20% of a division or 40 ms. **Random** spaces cuts at least length ÷ (4 × count) apart. Automatic chops are at least min(25 ms, length/64) long.

Loop BPM

The loop is assumed to be a power-of-two number of 4/4 beats (1, 2, 4 … 256), and the length whose tempo lands in 80–160 BPM is chosen. A 4-second file reads as 120 BPM (8 beats). If it's off by double or half, drag LOOP BPM.

Clock

Host: step = FL's quarter-note position × steps per quarter, looping from bar 1. A jump in FL (loop, click, tempo change) re-anchors without a stray hit. Internal: starts at step 1 on PLAY. Output is identical at any buffer size (tested at 64, 257 and 1,024 frames).

Sound

A 32-tap Blackman-windowed sinc with 64 sub-sample phases handles a different sample rate, step pitch and Tempo Fit. Reading faster than 1× switches to one of 81 narrower filters to stop aliasing. At matching rates with no pitch, samples are read directly with no filtering at all. Every voice fades in and out over 2 ms, and output is smoothed over 8 ms.

Real-time safety & CPU

The audio thread never allocates, locks or reads files. Edits publish a new snapshot, guarded by a hazard pointer. The editor polls at 30 Hz. The waveform and ribbon are cached images, so during playback only the playhead and step outline redraw. Debug and Release are both -O3. An unoptimised build says so in the footer.

Saved with your project

The decoded sample (so projects open even if the file moved), the mode and cuts, all four banks, randomizer settings and seed, and every parameter. That's about 23 MB per minute of 48 kHz stereo. A check marker rejects damaged data, and v0.1.0 sessions still open.

CHAPTER 10

Recipes & fixes

Flip a loop, keep the groove

GRID → 16 (or BEAT → 1/16), then IN ORDER. Turn on TEMPO FIT if tempos differ. Leave only **SWAP** on, set CHAOS to 20–35%, and roll until a swap catches your ear. The readout tells you what moved.

Chop a drum break

TRANSIENT at sensitivity 60–75, MAX 16. SWAP + RHYTHM + STUTTER at 50%. RATE 1/16, SWING 15–25%.

Vocal chops

TRANSIENT at low sensitivity, or RANDOM → 8. SWAP + PITCH + REVERSE at 40%. Pull gates down for stabs.

SYMPTOM	FIX
No sound	Press PLAY, or turn on HOST SYNC and start FL. Check the bank isn't all rests, then check OUTPUT and the mixer track.
Sound stops when FL is stopped	FL's *Smart disable* can sleep idle generators. Turn it off in the plugin wrapper's options menu (▼, top-left).
Everything sounds chipmunked	TEMPO FIT is on and LOOP BPM is wrong. Drag LOOP BPM to the real tempo, often double or half the detected value.
Clicks at chop starts	Turn ZERO-X on. In Manual, move the cut to a quiet point.
Beat chops land in odd places	Fix LOOP BPM first. Beat mode divides by it.
macOS says the plug-in is damaged	Run the `xattr -dr com.apple.quarantine` lines in INSTALL.txt (for CI downloads).

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